

PCCST501 – COMPUTER NETWORKS

Assignment 2 | Module 2 – Transport Layer & Network Layer

Reliable Communication & Routing Investigation

Course Outcome: CO2 **Due Date:** 03 AUGUST 2026

Study TCP/UDP communication and network-layer routing concepts through a combination of conceptual analysis, packet investigation, and routing-table work.

Instructions

1. Answer all questions. 2. Use diagrams wherever appropriate. 3. Cite any external material used. 4. Submit the PDF report through the prescribed GitHub repository and Google Form. 5. Screenshots must be readable and relevant.

Assignment Questions

1. Compare TCP and UDP for a real-time video call, file transfer, DNS query, and online game. Justify the protocol selected for each case.
2. Draw and explain the TCP connection-establishment process using the three-way handshake. Explain the purpose of SYN, SYN-ACK and ACK.
3. Explain TCP flow control and congestion control. Distinguish the receiver window from congestion-related control.
4. Using a packet-capture tool such as Wireshark, capture one TCP connection and identify source/destination IP addresses, source/destination ports, sequence numbers, acknowledgement numbers, flags and window information.
5. Explain the purpose of the network layer and compare forwarding with routing using a suitable example.
6. Create a small routed topology with at least three routers. Configure static routes and record the routing tables before and after configuration.
7. Explain the difference between intra-domain and inter-domain routing. Give one example of each.
8. Prepare a short note on multicast routing and explain one practical application.
9. Investigate IPv4/IPv6 transition requirements and explain why next-generation IP addressing is needed.
10. Prepare a one-page case study on QoS for a campus network carrying voice, video and ordinary data traffic.

Expected Deliverables

- PDF report
- Network topology diagram
- Packet-capture screenshots
- Routing-table screenshots
- README.md with conclusions

Suggested Evaluation

Criteria	Marks
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Correctness & Technical Content	4
Documentation & Diagrams	2
GitHub Repository & Organization	2
Presentation / Viva	2
Total	10

Submission Note

Create or use the assignment repository provided by the course, include a clear README.md, organize evidence into folders where appropriate, and submit the repository link through the course submission form.